

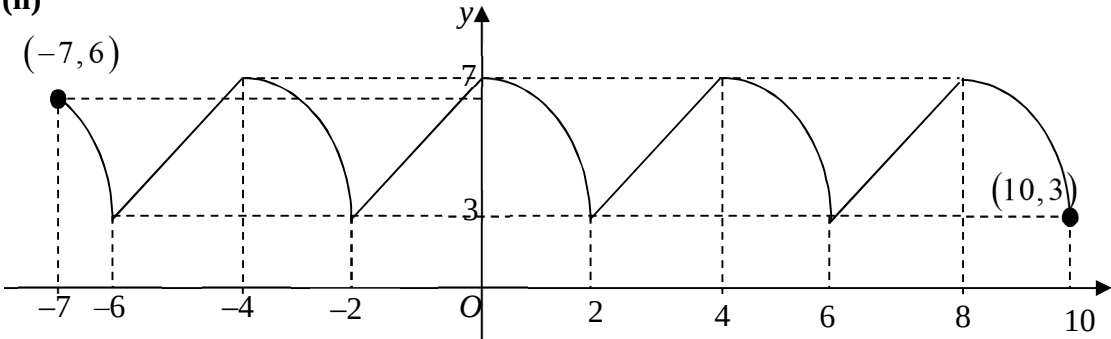
2009 'A' Level H2 Mathematics Paper 1 Suggested Solutions

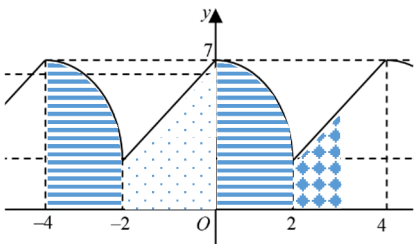
Qn 1	Suggested Solutions
1(i)	Let $u_n = An^2 + Bn + C$, where A, B and C are constants. $u_1 = A + B + C = 10$ $u_2 = 4A + 2B + C = 6$ $u_3 = 9A + 3B + C = 5$ Using GC, $A = \frac{3}{2}, B = -\frac{17}{2}, C = 17$ So, $u_n = \frac{3}{2}n^2 - \frac{17}{2}n + 17$ Be clear on how to write the general form for • Quadratic Polynomial • Cubic Polynomial • Quartic Polynomial
(ii)	$u_n = \frac{3}{2}n^2 - \frac{17}{2}n + 17 > 100$ $\frac{3}{2}n^2 - \frac{17}{2}n + 17 - 100 > 0$ $\begin{array}{c} + \quad \quad \quad - \quad \quad \quad + \\ \circ \quad \quad \quad \circ \\ -5.1266 \quad \quad \quad 10.793 \end{array}$ Using GC, $n > 10.793$ or $n < -5.1266$ Since $n \in \mathbb{Z}^+$, the set of values of n is $\{n \in \mathbb{Z}^+ : n \geq 11\}$. Set of values

Qn 2	Suggested Solutions
2	Include modulus for ln function to be defined Replace x by px, thus after integration remember to put in $\frac{1}{p}$ $\int_0^1 \frac{1}{4-x^2} dx = \int_0^{\frac{1}{2p}} \frac{1}{\sqrt{1-p^2x^2}} dx$ $\frac{1}{4} \left[\ln \left \frac{2+x}{2-x} \right \right]_0^1 = \frac{1}{p} \left[\sin^{-1}(px) \right]_0^{\frac{1}{2p}}$ $\frac{1}{4} \ln 3 = \frac{1}{p} \times \frac{\pi}{6}$ $p = \frac{2\pi}{3 \ln 3}$ Note: (Alternative method for $\int_0^{\frac{1}{2p}} \frac{1}{\sqrt{1-p^2x^2}} dx$) $\int_0^{\frac{1}{2p}} \frac{1}{\sqrt{1-p^2x^2}} dx$ $= \int_0^{\frac{1}{2p}} \frac{1}{\sqrt{p^2 \left(\frac{1}{p^2} - x^2 \right)}} dx$ $= \int_0^{\frac{1}{2p}} \frac{1}{p \sqrt{\left(\frac{1}{p^2} - x^2 \right)}} dx$ $= \left[\frac{1}{p} \sin^{-1} \left(\frac{x}{\frac{1}{p}} \right) \right]_0^{\frac{1}{2p}} = \frac{1}{p} \left[\sin^{-1}(px) \right]_0^{\frac{1}{2p}}$

Qn 3	Suggested Solutions
3(i)	$\frac{1}{n-1} - \frac{2}{n} + \frac{1}{n+1}$ $= \frac{n(n+1) - 2(n-1)(n+1) + (n-1)n}{(n-1)n(n+1)}$ $= \frac{n^2 + n - 2n^2 + 2 + n^2 - n}{(n-1)n(n+1)}$ $= \frac{2}{n(n-1)(n+1)}$ $= \frac{2}{n^3 - n}$ $\therefore A = 2$
(ii)	$\frac{1}{n-1} - \frac{2}{n} + \frac{1}{n+1} = \frac{2}{n^3 - n}$ $\frac{1}{n-1} - \frac{2}{n} + \frac{1}{n+1} = \frac{2}{n^3 - n}$ $\sum_{r=2}^n \frac{1}{r^3 - r} = \frac{1}{2} \sum_{r=2}^n \left(\frac{1}{r-1} - \frac{2}{r} + \frac{1}{r+1} \right)$ $= \frac{1}{2} \left[\frac{1}{1} - \frac{2}{2} + \frac{1}{3} + \frac{1}{2} - \frac{2}{3} + \frac{1}{4} + \frac{1}{3} - \frac{2}{4} + \frac{1}{5} + \dots \right]$ $+ \frac{1}{n-3} - \frac{2}{n-2} + \frac{1}{n-1}$ $+ \frac{1}{n-2} - \frac{2}{n-1} + \frac{1}{n}$ $+ \frac{1}{n-1} - \frac{2}{n} + \frac{1}{n+1}$ $= \frac{1}{2} \left(\frac{1}{2} + \frac{1}{n} - \frac{2}{n} + \frac{1}{n+1} \right)$ $= \frac{1}{2} \left(\frac{1}{2} - \frac{1}{n} + \frac{1}{n+1} \right)$ MOD can be applied to more than 2 terms and arrange the denominator in ascending/descending order for easy identification of cancellation of terms.
(iii)	This is the reason of convergence. As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$, $\frac{1}{n+1} \rightarrow 0$, $\sum_{r=2}^n \frac{1}{r^3 - r} \rightarrow \frac{1}{2} \left(\frac{1}{2} - 0 + 0 \right) = \frac{1}{4}$

Qn 3	Suggested Solutions
	Therefore, the series converges and $\sum_{r=2}^{\infty} \frac{1}{r^3 - r} = \frac{1}{4}$.

Qn 4	Suggested Solutions
4(i)	$f(x) = f(x+4)$ $f(45)$ $= f(41+4) = f(41)$ $= f(37+4) = f(37)$ $= \dots$ $= f(5)$ $= f(1) = 7 - (1)^2 = 6$ Similarly, $f(27) = f(23) = f(19) = \dots = f(3) = 2(3) - 1 = 5$ $\therefore f(27) + f(45)$ $= f(3) + f(1)$ $= 5 + 6$ $= 11$
(ii)	 The graph shows a periodic function $f(x)$ with a period of 4. The function is defined on the interval $[-7, 10]$. The graph consists of linear segments and semi-circular arcs. The linear segments are defined by $y = x + 6$ for $x \in [-6, -4]$, $[-2, 0]$, $[2, 4]$, and $[6, 8]$. The semi-circular arcs are defined by $y = -x^2 + 12x - 48$ for $x \in [-4, -2]$, $[0, 2]$, $[4, 6]$, and $[6, 8]$. The graph starts at a solid dot at $(-7, 6)$ and ends at a solid dot at $(10, 3)$. Dashed lines indicate the periodic structure and key points.

(iii)	$\int_{-4}^3 f(x) dx$  $= \int_{-4}^{-2} f(x) dx + \int_{-2}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^3 f(x) dx$ $= 2 \int_0^2 7 - x^2 dx + \int_2^4 2x - 1 dx + \int_2^3 2x - 1 dx$ $= 2 \left[7x - \frac{x^3}{3} \right]_0^2 + \left[x^2 - x \right]_2^4 + \left[x^2 - x \right]_2^3$ $= 2 \left(14 - \frac{8}{3} \right) + (12 - 2) + [6 - 2]$ $= 36 \frac{2}{3}$ Possible to use the area of trapezium $\frac{1}{2}(3+5)(1) = 4$ Possible to use the area of trapezium. $\frac{1}{2}(3+7)(2) = 10$ Note: It would be wrong to find $\int_{-4}^{-2} 7 - x^2 dx$ as the under the curve from $x = -4$ to $x = -2$ because $f(x) \neq 7 - x^2$ in this range.	Note:
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Qn 5 Out of syllabus (9758)

Qn 6	Suggested Solutions
6(i)	C_1 has asymptotes $x = -2$ and $y = 1$. Axial intercepts are $(0, -1)$ and $(2, 0)$. $C_2: \frac{x^2}{(\sqrt{6})^2} + \frac{y^2}{(\sqrt{3})^2} = 1$ C_2 is an ellipse with centre at origin and x-radius $\sqrt{6}$ and y-radius $\sqrt{3}$. Axial intercepts are $(0, \sqrt{3})$, $(0, -\sqrt{3})$, $(\sqrt{6}, 0)$ and $(-\sqrt{6}, 0)$.
(ii)	$C_1: y = \frac{x-2}{x+2} \Rightarrow y^2 = \left(\frac{x-2}{x+2}\right)^2$ $C_2: \frac{x^2}{6} + \frac{y^2}{3} = 1 \Rightarrow y^2 = \frac{1}{2}(6-x^2)$ The point of intersection between C_1 and C_2 is $\left(\frac{x-2}{x+2}\right)^2 = \frac{1}{2}(6-x^2)$ $2(x-2)^2 = (x+2)^2(6-x^2)$
(iii)	$x = -0.515$ or $x = 2.45$ Using part (ii), we could use the GC to solve the point of intersection between $2(x-2)^2$ and $(x+2)^2(6-x^2)$

Qn 7	Suggested Solutions
7(i)	Sub $y = e^{\cos x}$ to simplify derivatives. $f(x) = y = e^{\cos x}$ $f'(x) = -\sin x \cdot e^{\cos x} = -y \sin x$ $f''(x) = -y \cos x - \frac{dy}{dx} \sin x$ $f(0) = e^{\cos 0} = e$ $f'(0) = -e \sin 0 = 0$ $f''(0) = -e \cos 0 = -e$ Found in MF 26 By Maclaurin's expansion, $f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots$ $= e + 0 + \frac{x^2}{2!}(-e) + \dots$ $= e - \frac{e}{2}x^2 + \dots$ Express in the form $(1+x)^n$ before applying binomial theorem.
(ii)	Remember to bring in the power as well. $\frac{1}{a+bx^2} = (a+bx^2)^{-1}$ $= a^{-1} \left(1 + \frac{b}{a}x^2 \right)^{-1}$ $= a^{-1} \left(1 + (-1) \left(\frac{b}{a}x^2 \right) + \dots \right)$ $= a^{-1} \left(1 - \frac{b}{a}x^2 + \dots \right)$ $= \frac{1}{a} - \frac{b}{a^2}x^2 + \dots$ Comparing the constant: $\frac{1}{a} = e \Rightarrow a = \frac{1}{e}$ Comparing the coefficient of x^2: $-\frac{e}{2} = -\frac{b}{a^2} \Rightarrow b = \frac{a^2 e}{2} = \frac{1}{2e}$

Qn 8	Suggested Solutions
8(i)	$T_1 = a = 20$ $T_{25} = ar^{24} = 5$ $\Rightarrow r^{24} = \frac{1}{4} \Rightarrow r = \left(\frac{1}{4}\right)^{\frac{1}{24}}$ Since $r < 1$ (condition for sum to infinity to exist), S_∞ exists. $S_\infty = \frac{a}{1-r} = \frac{20}{1 - \left(\frac{1}{4}\right)^{\frac{1}{24}}} = 356.34 < 357$ $\therefore$ The total length of all the bars must be less than 357 cm, no matter how many bars there are. No matter how many bars $\Rightarrow$ consider as n tends towards infinity, thus we compute
(ii)	$L = S_{25} = \frac{20 \left(1 - \left(\frac{1}{4} \right)^{\frac{25}{24}} \right)}{1 - \left(\frac{1}{4} \right)^{\frac{1}{24}}} = 272.26 \approx 272 \text{ cm (3 s.f.)}$ $\text{Length of the 13th bar} = T_{13} = 20 \left(\frac{1}{4} \right)^{\frac{12}{24}} = 10 \text{ cm}$
	Total length of the 25 bars = 272.26 $\frac{25}{2} [u_1 + u_{25}] = 272.26$ $u_1 + 5 = 21.7808$ $u_1 = 16.7808$ $= 16.8 \text{ (3 s.f.)}$ Length of the longest bar = $u_1 = 16.8 \text{ cm (3 s.f.)}$ Since the metal bars are of decreasing lengths, the longest bar will be the first bar. $u_{25} = 5$ $u_1 + 24d = 5$ $d = \frac{5 - 16.7808}{24}$ $d = -0.491 \text{ (3 s.f.)}$

Qn 9 Out of syllabus (9758)

Qn	Suggested Solutions
10(i)	Let θ denote the angle between p_1 and p_2. $\cos \theta = \frac{\begin{vmatrix} 2 \\ 1 \\ 3 \end{vmatrix} \cdot \begin{vmatrix} -1 \\ 2 \\ 1 \end{vmatrix}}{\begin{vmatrix} 2 \\ 1 \\ 3 \end{vmatrix} \begin{vmatrix} -1 \\ 2 \\ 1 \end{vmatrix}} = \frac{3}{\sqrt{14}\sqrt{6}}$ $\theta = 70.9^\circ$ Qn asked for acute angle, thus we should insert modulus to ensure that the angle obtained is always acute.
(ii)	The equations of p_1 and p_2 are $2x + y + 3z = 1$ $-x + 2y + z = 2$ Using GC $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 - z \\ 1 - z \\ z \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}.$ The equation of line of intersection is $l: \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$ Do not forget to add in these.
(iii)	$2x + y + 3z - 1 + k(-x + 2y + z - 2) = 0$ $\Rightarrow (2 - k)x + (1 + 2k)y + (3 + k)z = 1 + 2k$ Equation of p_3 can be written as $\mathbf{r} \cdot \begin{pmatrix} 2 - k \\ 1 + 2k \\ 3 + k \end{pmatrix} = 1 + 2k$. Rearrange the given form into the scalar product form first. Method 1: Subst l into p_3 $\text{LHS} = \begin{pmatrix} -\lambda \\ 1 - \lambda \\ \lambda \end{pmatrix} \cdot \begin{pmatrix} 2 - k \\ 1 + 2k \\ 3 + k \end{pmatrix}$ $= -\lambda(2 - k) + (1 - \lambda)(1 + 2k) + \lambda(3 + k)$ $= -2\lambda + \lambda k + 1 + 2k - \lambda - 2k\lambda + 3\lambda + \lambda k$ $= 1 + 2k = \text{RHS}$ Since this is a showing question, please do not skip steps and show out every single step.

Method 2:

$$\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2-k \\ 1+2k \\ 3+k \end{pmatrix} = -2+k-1-2k+3+k=0$$

so the line l is parallel to p_3 for any constant k .

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2-k \\ 1+2k \\ 3+k \end{pmatrix} = 1+2k$$

This proves that a point on line l satisfies the equation of p_3 .
Thus, the point lies in p_3 .

so the point with position vector $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ lies on p_3 for any constant k .

Since l is parallel to p_3 and a point that lies on l also lies in p_3 , thus l lies in p_3 for any constant k .

‘Hence’ Method:

Since we need to find a cartesian equation of the plane in which both l and the point $(2, 3, 4)$ lie in, we can link to the earlier part.

Since $(2, 3, 4)$ lies in the plane, we can find a specific value of k such that p_3 contains $(2, 3, 4)$.

$$\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 2-k \\ 1+2k \\ 3+k \end{pmatrix} = 1+2k$$

$$\Rightarrow 8k+19=1+2k$$

$$\Rightarrow k=-3$$

Thus the equation of the plane in which both l and the point $(2, 3, 4)$ lie is

$$\mathbf{r} \cdot \begin{pmatrix} 2-(-3) \\ 1+2(-3) \\ 3+(-3) \end{pmatrix} = 1+2(-3). \quad \text{i.e., } \mathbf{r} \cdot \begin{pmatrix} 5 \\ -5 \\ 0 \end{pmatrix} = -5$$

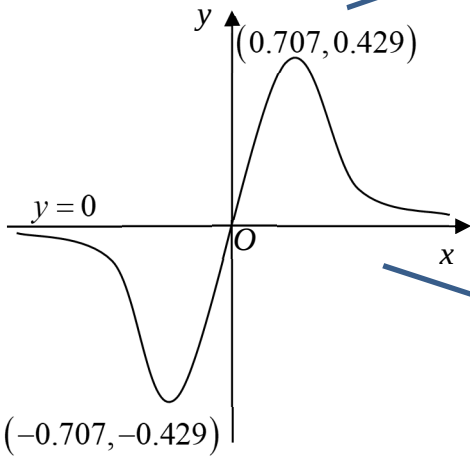
The cartesian equation is $x-y=-1$ $z \in \mathbb{R}$.

Since the dot product between the direction of l and the normal vector of p_3 is 0, the line l is parallel to p_3 . However, it is insufficient to conclude that the line lies in the plane. We need a second condition.

2 conditions are required if we want to prove that a line lies in a plane.

1. The line is parallel to plane
2. A point on the line lies on the plane.

This must be written, otherwise the equation becomes a line and not a plane.

Qn	Suggested Solutions
11(i)	 We may leave the coordinates of the turning points in 3 s.f. Show clearly the asymptote $y = 0$ as the graph approaches the x-axis.
(ii)	$f(x) = xe^{-x^2}$ $f'(x) = e^{-x^2} + xe^{-x^2}(-2x) = e^{-x^2} - 2x^2e^{-x^2}$ $e^{-x^2} - 2x^2e^{-x^2} = 0$ $e^{-x^2}(1 - 2x^2) = 0$ $e^{-x^2} = 0 \quad \text{or} \quad x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}}$ (N.A $\because e^{-x^2} > 0$) When $x = \frac{1}{\sqrt{2}}, y = \frac{1}{\sqrt{2}}e^{-\left(\frac{1}{\sqrt{2}}\right)^2} = \frac{1}{\sqrt{2}}e^{-\frac{1}{2}} = \frac{1}{\sqrt{2}e^{\frac{1}{2}}} = \frac{1}{\sqrt{2e}}$ When $x = -\frac{1}{\sqrt{2}}, y = -\frac{1}{\sqrt{2}}e^{-\left(-\frac{1}{\sqrt{2}}\right)^2} = -\frac{1}{\sqrt{2}}e^{-\frac{1}{2}} = -\frac{1}{\sqrt{2}e^{\frac{1}{2}}} = -\frac{1}{\sqrt{2e}}$ Coordinates of the turning points are $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2e}}\right)$ and $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2e}}\right)$

(iii)	$x = 0 \Rightarrow u = 0$ $x = n \Rightarrow u = n^2$ $u = x^2 \Rightarrow x = \sqrt{u} \Rightarrow \frac{dx}{du} = \frac{1}{2\sqrt{u}}$ $\int_0^n f(x) dx = \int_0^n x e^{-x^2} dx$ $= \int_0^{n^2} \sqrt{u} e^{-u} \frac{1}{2\sqrt{u}} du$ $= \frac{1}{2} \int_0^{n^2} e^{-u} du$ $= \frac{1}{2} \left[-e^{-u} \right]_0^{n^2}$ $= \frac{1}{2} (-e^{-n^2} + e^0)$ $= \frac{1}{2} (1 - e^{-n^2})$ Area of region between C and the positive x-axis $= \lim_{n \rightarrow \infty} \int_0^n f(x) dx = \frac{1}{2} (1 - 0) = 0.5$
(iv)	$\int_{-2}^2 f(x) dx = 2 \int_0^2 f(x) dx$ $= 2 \times \frac{1}{2} (1 - e^{-2^2})$ $= 1 - e^{-4}$ Using result in part (iii) As $n \rightarrow \infty$, $e^{-n^2} \rightarrow 0$
(v)	Vol of revolution $= \pi \int_0^1 (f(x))^2 dx$ $= \pi \int_0^1 (x e^{-x^2})^2 dx$ $= 0.11570\pi$ $= 0.36349$ $= 0.363 \quad (3 \text{ s.f.})$ $\int_{-2}^2 f(x) dx$ is the same as finding the area bounded by $f(x)$ and x-axis from $x = -2$ to $x = 2$. $f(x)$ is symmetrical about y-axis. So the area under the graph from $x = -2$ to $x = 0$ is the same as the area under the graph from $x = 0$ to $x = 2$.